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A Tutorial on What to Do With Skewness, Kurtosis, and Outliers: New Insights to Help Scholars Conduct and Defend Their Research

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ABSTRACT

This research examines the effect of nonnormal data on several frequently utilized statistical models. Behavioral scholars often encounter nonnormal and potentially problematic data such as outliers. In the absence of clear guidance, scholars generally report taking some compensatory action, including the deletion of data, data transformations, or alternative statistics. This research addresses the question, how much do such data actually affect results? Three sets of simulations investigate the effects of skewness, kurtosis, and outliers. Our findings show that some patterns of nonnormal data are perhaps not as problematic as one might anticipate. Many results are affected but in the statistically conservative direction (i.e., less likely to produce significance). In particular, data that were skewed, or showed kurtosis or univariate outliers did not lead to inappropriate statistical results, and, therefore, scholars may proceed with their analyses. Yet there are boundary conditions; specifically, we found that multivariate outliers could create biased results, so for such data, scholars need to be particularly transparent and vigilant in their data handling, posting data sets with and without any deleted or transformed data points.

1 | Introduction

Behavioral scholars conducting empirical research typically begin their investigations by calculating simple descriptive statistics to verify the basic qualities of the data: Does everything look sensible? If so, then they can proceed to model fitting and theory testing. If not—what should researchers do? Similarly, what should reviewers ask about data processing when they assess manuscripts? With these questions, scholars and reviewers are not necessarily raising issues of data mismanagement; rather, there is some guidance (e.g., Tukey 1977) but not a great deal of consensus about how to handle several types of data (hence, we present the suggested solutions shortly). For example, should outliers always be deleted? If they are not, are the results biased? When a paper reports that results

were the same with and without deleted observations, why were those data deleted?

Given that scholars are trying to do the right thing with their data in their research discoveries, some guidance may be helpful. If analyses are affected by nonnormal data, it would help to have norms regarding data transformations or the use of nonparametric methods (Aguinis, Gottfredson, and Joo 2013; Dixon 1953; Doksum and Wong 1983; R. J. Lawrence 1980; Wilcox 2017) or the broader generalized linear model (McCullagh and Nelder 2019). Alternatively, if data analyses are not greatly affected by nonnormal data, or if the data do not deviate from normality by much (and we will investigate “how much”), then perhaps data deletions would be unnecessary or even discouraged for purposes of research transparency.

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To be more precise, when we reference “nonnormal data,” we do not mean simply “odd-looking data” (though nonnormal data can certainly look odd). We mean that the data do not follow the distribution of normality, which is assumed in the hypothesis tests of parameters in many popular statistical models. Specifically, in our studies, we examine three major ways that a data distribution can appear to be nonnormal. A data distribution can exhibit (1) skewness (points to the left or right), (2) kurtosis (flat or peaked), or (3) it may contain outliers. These three data qualities are conceptually independent of each other (e.g., a data set can display kurtosis or skewness without outliers), just as means and variances are independent of each other for the normal distribution, hence our motivation to run three separate simulations to isolate unconfounded effects. There are other data qualities that we do not examine, including issues of range restriction per se nor discrete data.) We will elaborate on the three different data qualities—skewness, kurtosis, and outliers—throughout the paper and across the three simulation studies, with precise definitions appearing at the beginning of each study and in the Appendix. We will also distinguish between data that may be outliers versus those that arise from errors, such as input or scanning errors.

If a data set differs from its theoretically assumed distribution, it is important to know the extent to which the model results may be affected by possibly violated assumptions (Rubel and Naik 2017). In behavioral literature, scholars seem aware of and attuned to screening for outliers, yet data sets can also deviate from perfect standard normality by displaying skewness or kurtosis, and neither of these qualities seems to be checked or reported in journal articles as frequently as a search and reporting of outliers. Given that outliers, skewness, and kurtosis are distinct data characteristics, one question is whether any one of those data properties might be problematic. Thus, we believe it to be helpful to study each of these nonnormalities. In extending the literature, we examine each of these types of data, investigating their possible effects on correlations and regression models, per Figure 1.

In this research, we investigate these three forms of nonnormal and potentially problematic data to assess their effects on different models. That is different models, or different parts of models, may be differentially sensitive to nonnormal data. For example, if we find that a correlation is unaffected by modest outliers, but a regression model with multiple predictors becomes biased, then deletions or transformation or other efforts would not be necessary for correlations, yet they would be more critical in regressions. Our research adds to the literature on robustness (cf., in psychology, biometrics, statistics, social sciences) by offering a more complete assessment of forms of

nonnormality and the impact on various statistical analyses (Bakker and Wicherts 2014; Bishara and Hittner 2012; Box and Watson 1962; Hogg 1979; Wilcox 2023). The literature on robustness suggests that, *theoretically*, nonnormal data would be problematic because such data violate distributional assumptions of statistical models. Our research tests whether those statistical theoretical predictions hold true; that is, *empirically*, what do results look like? Is there bias in a regression coefficient or statistical inefficiency regarding standard errors, and so forth?

1.1 | Understanding the Problem and Corrective Actions

To get more specific and set the data problem in context, we begin by more precisely defining what it means for data to be nonnormal or potentially problematic. In real, empirical data, it can certainly be the case where “most” of the data set seems to follow a normal distribution, at least roughly, but sometimes there may be one or more data points that stand out as different. Such data may arise as errors in data input or they may be outliers (defined precisely at the start of each study and in the Appendix). In this research, we are not addressing *errors* that require resolution. In situations with blatant and demonstrable errors, presumably scholarly norms would find it acceptable for researchers to rectify the error, and clean the data set, likely by deleting that observation (Karen 2021; Wilcox 2017). Furthermore, such errors are easily documented.

In contrast, sometimes data can look nonnormal even without being an error. (Of course, some data might follow alternative distributional forms, where the research team will be using proper statistical models that indeed expect those forms, such as Poisson distributions or negative binomials, but the models examined in this research focus on the assumptions of the normal distribution as a common distribution, hence, if the normal distribution is assumed, we examine whether it is problematic when the data do not seem to conform to the assumptions.) These nonnormal or unusual data are the focus of this research. Many articles in recent issues of nearly any empirical journal offer examples of researchers dealing with nonnormal data, such as data that exhibit skewness or kurtosis or a data set with outliers (e.g., consider these citations from the management literature, psychology, consumer behavior, marketing, consumer psychology, and computational statistics: Aguinis, Gottfredson, and Joo 2013; Bakker and Wicherts 2014; Howard et al. 2022; Mittelman, Gonçalves, and Andrade 2020; Moon, Tuli, and Mukherjee 2023; Troll et al. 2023; Yu et al. 2019; K. H. Yuan and Zhong 2013). When nonnormality is a function of outliers, the potentially problematic data are often addressed by deletion, yet

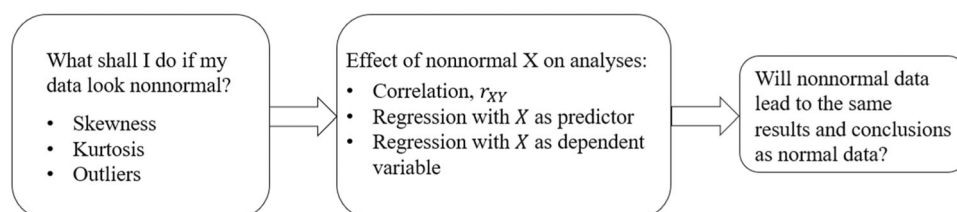


FIGURE 1 | Overview of three sets of simulation studies.

without providing extensive explanations or justification. Similarly, little or no rationale is typically offered when researchers address skewness or kurtosis with transformations (all of which we describe shortly). Our research is not intended to question a researcher's remedy actions; indeed, the aforementioned articles follow good practice. Yet in today's scholarly environment, the deletion of any data can raise doubts to readers and reviewers who might become suspicious that the researchers were culling the data set toward support of their hypotheses, even when that is not the researchers' intention.

Given the prevalence of nonnormal data, scholars have offered strategies for corrective analyses (cf., K. D. Lawrence 2020; Stahel and Weisberg 1991). Data may be modified, or deleted ("trimmed") or "Winsorized" (extremes are trimmed, as defined in the Appendix and described next). Let's discuss each.

First, scholars may transform the variable to appear more normal-like (Cain, Zhang, and Yuan 2017; Ratcliff 1993). (And of course note that a transformation modifies all the data, not just outliers, to resemble more closely an assumed normal distribution.) Natural logs and square root transformations are often suggested to minimize large data values, and while they succeed in part, they typically do not bring the data to standard normal parameters sufficiently (Games and Lucas 1966; R. J. Lawrence 1980; Lo and Andrews 2015; Rasmussen 1989). For example, in the Appendix, we show distributions of skew 0.5, 0.75, and 1.0 transformed by square roots yield only slightly lesser skewness of 0.31, 0.63, and 0.87. Natural log transformations were somewhat better, yielding 0.06, 0.44, and 0.69, but these are still nonzero (please see the Appendix for more detail and Anglim et al. [2020] for an example). Second, scholars have recommended the use of robust statistics—indices created to be less influenced by nonnormal data, such as the use of medians, Mann–Whitney Wilcoxon tests on ranks, Winsorizing observations, M- or W-estimators based on maximum likelihood and differential weighting (these alternatives are explained in the Appendix; Bakker and Wicherts 2014; Boos and Stefanski 2015; Fabozzi et al. 2014; Hettmansperger and McKean 2010; Huber 1964; Pollet and van der Meij 2017; Yu et al. 2019; Y. Yuan and MacKinnon 2014). Some scholars (e.g., Bakker and Wicherts 2014) warn against removing outliers, for example, and recommend trimmed means or nonparametric tests. Trimming data removes observations, whereas Winsorizing (removing extreme values, defined in the Appendix) lessens their absolute value, but there are no norms to guide scholars as to the percentage of observations to transform. Using medians implies standard statistical tests and distributions cannot be applied as standard errors and least squares estimation no longer hold, and while Mann–Whitney Wilcoxon or any tests on ranks can be legitimate, scholars usually gather interval or ratio data because they are curious about the strengths of effects and relationships which are now obviated in ranks. Thus these alternatives do not provide perfect solutions either. Further, these indices appear infrequently in the literature, perhaps due to lesser familiarity and also perhaps due to little provision of meaningful norms for reporting and interpretation (please see the Appendix for more on these indices).

Thus, our analyses are intended to be helpful, dealing with the statistics and the philosophy of scientific discoveries. A simple,

legitimate question arises: is it even necessary to delete *any* data? Our research addresses the question: what should researchers do with data points like outliers—delete them or leave the data as they are and proceed with analyses? Our findings mostly offer the assurance that such data have negligible impact on common analyses and in fact make the results more statistically conservative. That is, the results bend in the traditional direction to support scientific progress, hypothesis and theory testing, and publishing, compared to a statistically liberal direction which could contribute to problems of replication or retraction requests. The conservative results are not uniform, however, which we will document clearly when data yield problematic findings. Specifically, our findings show that data that were skewed or showed kurtosis or univariate outliers were not problematic, and scholars may proceed with their analyses as results were affected in the statistically conservative direction. However, data with multivariate outliers were problematic, yielding biased results, and scholars need to document any data adjustments they invoke.

These research questions matter because numerous statistical models assume specific distributional forms, often, normality. For example, in the ubiquitous regression model, $y = b_0 + b_1X_1 + b_2X_2 + \epsilon$, X or y need not be normal but the errors are assumed to be normal for valid use of significance tests (Allison 1999; Kennedy 2008; Kleinbaum et al. 2013; Montgomery, Peck, and Vining 2012). That is, normality is not required in the least squares estimation of regression coefficients, but it is necessary in statistical tests of their significance, which is a fundamental part of most analyses and reporting. Fortunately, normal distributions are indeed pervasive, from people's heights (some short, some tall, most average), to levels of intellect (some are smart, some are not, most are average), to attitudes toward many stimuli such as electric vehicles (some positive, some negative, most moderate), thus the assumption of normality is often quite reasonable. Normal distributions are not universal, of course, for example, household or country wealth tend to be positively skewed.

In this research, we investigate whether nonnormal and potentially problematic data matter. Theoretically, researchers "know" that an outlier can affect a mean. For example, in comparing two small data sets (1, 2, 3, 4, 5) and (1, 2, 3, 4, 10), the mean for the first is 3, and the mean for the second is 4. If those sample sizes were expanded from $N=5$ to $N=30$ (a modest increase per the Central Limit Theorem), the effect of an outlying data point is already much reduced. Specifically, if the (1, 2, 3, 4, 5) data set is replicated six times, the mean would remain 3, but if one of those 5's were changed to 10, the mean changes merely to 3.1667. Is that change significant? In the first example, with small $N=5$, the mean increases but so does the standard error (0.79 to 1.77), thus compensating in a t -statistic ratio trending down, toward being more conservative ($3/0.79 = 3.80$ to $4/1.77 = 2.26$; we say more about this shortly). Therefore, while *theoretically* we can all agree that nonnormal data appear worrisome, in this research, we examine the extent to which *empirically* such data actually have troubling effects.

To understand the possible effects of nonnormal and potentially problematic data, this paper is organized as follows. We present three sets of simulation studies wherein we examine skewness,

kurtosis, and outliers. We examine their effects on several frequently employed statistical models. We close with recommendations for researchers on how to effectively deal with skewness, kurtosis, or outliers giving rise to nonnormal data. The Appendix provides a brief tutorial refresher on definitions of skewness, kurtosis, and outliers. The Supporting Information S1: [Web Appendix](#) provides additional analyses, showing confirmation of our findings across several real-world data sets as well.

2 | Study I: Skewness

Studies I.a.–I.g. examine the effect of skewed data on the results of several frequently utilized statistical models. Skewness is a function of the third moment of the normal distribution, X^3 ; $skew = \frac{\sum(x_i - \bar{x})^3}{(n)\sigma^3}$ (Aron, Coups, and Aron 2012; Cribbie et al. 2012; Ratcliff 1993; Sharma, Durvasula, and Dillon 1989; Wilcox 2021; for more detail, please see the Appendix). It captures the symmetry or asymmetry of the distribution. Skewness is positive if there are some large observations in the right tail (and the mean is greater than the median), and negative if such observations appear in the left tail (see examples in the Appendix). Skewness can be pervasive in continuous measures (e.g., sales) but can also manifest in Likert scales; for example, if the majority of responses are 1's and 2's, but there exist several 7's or even 6's, skewness can be nonzero (cf., Liddell and Kruschke 2018; Olsson 1979). In perfect standard normal distributions, skewness equals zero, and statistical packages like SAS and SPSS routinely provide skewness indices along with means and standard deviations.

In Studies I.a.–I.g., standard normal distributions were generated using SAS's `randnormal` function for three variables (x , y , z), for sample sizes $N = 30, 100, 500, 1000$. The distributions for variable x were transformed to skewness levels of 0.0, 0.5, 0.75, 1.0 (Fleishman 1978; Reinartz, Echambadi, and Chin 2002). Negative skews were not generated given the results would merely be reversed. Data can show greater skew indices but then kurtosis is also affected, and our studies aimed to isolate the effects (e.g., Cain, Zhang, and Yuan 2017). For each combination of sample size and skewness, we generated 1000 replications. For each of the $4 \times 4 \times 1000$ samples, we fit several frequently employed models:

1. We examined simple, pairwise correlations between x (skewed) and y (normal; Bishara and Hittner 2012; Wilcox 2023). The simple correlation is the basis upon which many other models are built, so if correlations are not robust, it would not be surprising to find problems in regressions as well. (Note also that the data were created to surpass the trivariate normality assumption, implying the univariate derived distributions would also start normal, before transformation to different levels of skewness.) Credé (2010) offers precedence in examining correlations, regressions, and the relationships between these manifestations of the general linear model and other variants.
2. We fit a multiple linear regression where skewed x was a predictor, and the normal variables served as a second

predictor (z) and dependent variable (y). The second predictor was included to allow multicollinearity to be more realistic than a single predictor regression. The regression models y , but the function form and shape of a predictor like x can modify the relationship between these two (or three) variables. Given the status of regression and the general linear model as one of the most dominant families of statistical models, understanding the regression model with nonnormal data should be important.

3. We fit regressions where skewed x was the dependent variable, and the normal variables y and z were predictors. It is plausible that a nonnormal error and variable would be less influential in its role as an independent or dependent variable, thus in testing both models, we compare the effect of violations of normality when x is an independent or dependent variable.

The effects of skewness were also examined on additional models: (a) regressions where the skewed x was a predictor (along with z , using y as a dependent variable), in a model with an interaction term, $x \times z$. Product terms are common for testing moderators or boundary conditions. (b) We fit binary logit models using skewed x (and normal z) as predictors, with a binary split on y as a dependent choice variable. (c) We fit mediation analyses via structural equations models, with the paths $x \rightarrow z$, $z \rightarrow y$, and $x \rightarrow y$; x being the skewed variable. Mediation analyses are frequently used to detect the process through which an independent variable x may affect a dependent variable y , through the mediator z . It is perhaps not surprising, given the relationships among these variants of the general linear model, that the results of these additional models were consistent with the results we present shortly, thus we focus on correlations and regressions with a skewed variable being the predictor or the dependent variable.

For each model, we examine the effects on three classes of statistics, assessing: (1) the unbiasedness of the parameter estimates (e.g., b_x in regression), (2) the efficiency of those estimates as reflected in their standard errors (e.g., $se(b_x)$), and (3) the overall model fit statistics (e.g., R^2). Next, we specify the designs of Studies I.a.–I.g.

3 | Study I Results

In Study I.a., the three variables were generated having modest intercorrelations ($\rho_{xy} = 0.3$, $\rho_{xz} = 0.3$, $\rho_{yz} = 0.3$). We reasoned that intercorrelations of 0 would be unusual in data, and results obtained thereby may be misleading. The x variable was then transformed to follow skewness levels of 0.0, 0.5, 0.75, 1.0 (and the manipulated skewness levels were confirmed by descriptive statistics yielding skewness values of 0.01, 0.48, 0.73, 0.98).

Table 1 indicates the pattern of significant results across the correlation estimates and the regressions. Most effects were significant, driven in part by power due to the size of the simulation study (e.g., 1000 replications in each cell), thus Table 1 also offers effect sizes as complementary information (η_p^2 is interpretable like R^2 , Cohen 1973). The effect sizes were negligible, often explaining less than 1% of the variance.

TABLE 1 | Results for skewness: *p*-values and effect sizes [Studies I.a.–I.g.].

	Study I.a.: r_{xy}'s 0.3	Study I.b.: r_{xy}'s 0.5	Study I.c.: r_{xy}'s 0.7	Study I.d.: r_{xy}'s 0.3, 0.5, 0.7	Study I.e.: greater skew 0, 1, 1.5, 1.75	Study I.f.: X and Y multivariate	Study I.g.: multivariate opposite
Correlations							
r_{xy}	$p < 0.0001$ $\eta_p^2 = 0.010$	$p < 0.0001$ $\eta_p^2 = 0.036$	$p < 0.0001$ $\eta_p^2 = 0.122$	$p < 0.0001$ $\eta_p^2 = 0.010$	$p < 0.0001$ $\eta_p^2 = 0.038$	$p < 0.0001$ $\eta_p^2 = 0.018$	$p < 0.0001$ $\eta_p^2 = 0.008$
t_{xy}	$p < 0.0001$ $\eta_p^2 = 0.033$	$p < 0.0001$ $\eta_p^2 = 0.113$	$p < 0.0001$ $\eta_p^2 = 0.323$	$p < 0.0001$ $\eta_p^2 = 0.033$	$p < 0.0001$ $\eta_p^2 = 0.122$	$p < 0.0001$ $\eta_p^2 = 0.059$	$p < 0.0001$ $\eta_p^2 = 0.027$
p_{xy}	$p = 0.003$ $\eta_p^2 = 0.001$	$p = 0.003$ $\eta_p^2 = 0.002$	$p < 0.0001$ $\eta_p^2 = 0.002$	$p = 0.003$ $\eta_p^2 = 0.001$	$p = 0.006$ $\eta_p^2 = 0.001$	$p < 0.0001$ $\eta_p^2 = 0.003$	$p = 0.0002$ $\eta_p^2 = 0.001$
Regression X predictor							
b_x	$p < 0.0001$ $\eta_p^2 = 0.006$	$p < 0.0001$ $\eta_p^2 = 0.023$	$p < 0.0001$ $\eta_p^2 = 0.084$	$p < 0.0001$ $\eta_p^2 = 0.002$	$p < 0.0001$ $\eta_p^2 = 0.022$	$p < 0.0001$ $\eta_p^2 = 0.008$	$p < 0.0001$ $\eta_p^2 = 0.004$
R^2	$p < 0.0001$ $\eta_p^2 = 0.003$	$p < 0.0001$ $\eta_p^2 = 0.006$	$p < 0.0001$ $\eta_p^2 = 0.016$	n.s.	$p < 0.0001$ $\eta_p^2 = 0.006$	$p < 0.0001$ $\eta_p^2 = 0.017$	$p < 0.0001$ $\eta_p^2 = 0.010$
std. error(x)	n.s.	$p = 0.0087$ $\eta_p^2 = 0.001$	$p < 0.0001$ $\eta_p^2 = 0.007$	$p < 0.0001$ $\eta_p^2 = 0.002$	$p < 0.0001$ $\eta_p^2 = 0.002$	n.s.	n.s.
Regression X dep var							
b_y	$p < 0.0001$ $\eta_p^2 = 0.004$	$p < 0.0001$ $\eta_p^2 = 0.007$	$p < 0.0001$ $\eta_p^2 = 0.010$	$p = 0.0003$ $\eta_p^2 = 0.001$	$p < 0.0001$ $\eta_p^2 = 0.020$	$p < 0.0001$ $\eta_p^2 = 0.007$	$p < 0.0001$ $\eta_p^2 = 0.004$
R^2	$p < 0.0001$ $\eta_p^2 = 0.020$	$p < 0.0001$ $\eta_p^2 = 0.065$	$p < 0.0001$ $\eta_p^2 = 0.193$	$p < 0.0001$ $\eta_p^2 = 0.046$	$p < 0.0001$ $\eta_p^2 = 0.058$	$p < 0.0001$ $\eta_p^2 = 0.021$	$p < 0.0001$ $\eta_p^2 = 0.015$
std. error(y)	$p = 0.0002$ $\eta_p^2 = 0.001$	$p = 0.0001$ $\eta_p^2 = 0.006$	$p < 0.0001$ $\eta_p^2 = 0.030b$	$p < 0.0001$ $\eta_p^2 = 0.004$	$p = 0.0003$ $\eta_p^2 = 0.001$	$p = 0.0269$ $\eta_p^2 = 0.001$	$p = 0.0016$ $\eta_p^2 = 0.001$

Note: All results across skew levels of 0, 0.5, 0.75, 1, were statistically conservative: with more skew, parameters and fits fall in size (e.g., r , t , b , R^2), and rise for standard errors. η_p^2 interpretable like R^2 (Cohen 1973). Results for Study I.g. were similarly conservative or modestly conservative (e.g., standard errors of 0.090, 0.090, 0.090, 0.091).

The effect of skewness on correlations was significant. The correlations drop from 0.298 (no skew) to 0.273 (skew = 1). The t -statistics drop with skewness from 5.46 to 4.98 and p -values rose from 0.062 to 0.072, all results being statistically conservative. The regression results were also statistically conservative. When skewed x is a predictor, the b -coefficients dropped (0.230–0.208) and R^2 's dropped (0.156–0.147). When skewed x is the dependent variable, b -coefficients decreased (0.229–0.213), as did R^2 's (0.158–0.134), and standard errors were steady or rose slightly (0.0899–0.090).

Study I.b. considers higher correlations; the intercorrelations of 0.3 in Study I.a. may have been too modest to yield compelling results. Study I.b. examines slightly greater intercorrelations ($\rho_{xy} = 0.5$, $\rho_{xz} = 0.5$, $\rho_{yz} = 0.5$). Once again, the patterns are statistically conservative. As skewness rises, the correlations drop (0.498 to 0.457), the t -statistics drop (10.01–8.96) and the p -values rise (0.009–0.015). In the regressions with x as a predictor, the b -values drop (0.332–0.291) and the R^2 values drop (0.344–0.328). The standard errors also decreased slightly (0.088–0.086), yet when scholars report regression results, they would presumably report whether the b -values were significant,

and in using the b -values with the standard errors, the respective t -values indeed drop (from 3.77 to 3.38). In the regressions with x as a dependent variable, the b -values drop (0.331–0.308), the R^2 's drop (0.346–0.293), the standard errors rise (0.087–0.091), for lower t -values (3.80–3.38).

Study I.c. examined intercorrelations high enough to indicate multicollinearity ($\rho_{xy} = 0.7$, $\rho_{xz} = 0.7$, $\rho_{yz} = 0.7$), and the pattern of statistically conservative findings continue. As skewness rises, the correlations drop (0.698–0.641), the t -statistics drop (17.01–14.57) and the p -values rise (0.0002–0.001). In the regressions with x as a predictor, the b -values drop (0.411–0.331) and the R^2 's drop (0.581–0.559). The standard errors also drop (0.085–0.081), but the respective t -values drop (from 4.84 to 4.09). In the regressions with x as a dependent variable, the b -values drop (0.409–0.381), the R^2 's drop (0.582–0.494), and the standard errors rise (0.084–0.093).

Study I.d. includes a mix of correlations ($\rho_{xy} = 0.3$, $\rho_{xz} = 0.5$, $\rho_{yz} = 0.7$). The findings are statistically conservative. As skewness rises, the correlations drop (0.298–0.273), the

t -statistics drop (5.46–4.98), and the p -values rise (0.062–0.072). In the regressions with x as a predictor, the b -values drop in magnitude (–0.069 to –0.058; the negative signs presumably being attributable to the greater correlation between the second predictor and the dependent variable, compared with x). The standard errors also drop slightly (0.076–0.074), but again the respective t -values drop (–0.908 to –0.784). In the regressions with x as a dependent variable, the b -values drop (–0.102 to –0.089), the R^2 's drop (0.271–0.228), and the standard errors rise (0.112–0.115).

Study I.e. extends the level of skewness. Fleishman (1978) provides transformative values from skewness of 0.25 to 1.75. We had previously capped skewness at 1.0 for two reasons. First, a skewness level of 1.0 is already visibly not normal in appearance or shape; it is asymmetric like a ski slope, as if a normal distribution were split down the middle (see the Appendix). Second, we were trying to examine the effects of skewness and kurtosis independently of each other, and as skewness rises (e.g., to 1.75), so does kurtosis (to 3.75), reflecting that as skewness gets more extreme, the distribution gets more peaked. We generated an anecdotal data set to push skewness (with the values of: five 1's, 2–7, 15, 25), and the skewness increased (i.e., 2.62) but so did kurtosis (7.13). (For our skewed values of 0.5, 0.75, and 1.0, the kurtosis values were negligible; 0.0, 0.0, 0.5, respectively.) Still, the question remained, “What about variables that are even more skewed?” so in Study I.e., we compare skewness levels of 0.0, 0.5, 1.0, 1.75 (the maximum possible for these transformations). These results also show a pattern of statistically conservative findings. As skewness rises, the correlations drop (0.298–0.243), t -statistics drop (5.46–4.40), and p -values rise (0.062–0.091). In the regressions with x as a predictor, the b -values drop (0.230–0.183), the R^2 's drop (0.156–0.142), and standard errors rise (0.090–0.093). In regressions with x as a dependent variable, the b -values drop (0.229–0.185), the R^2 's drop (0.158–0.112), and standard errors rise (0.090–0.091).

Study I.f. considers the multivariate question, “What happens if two variables are skewed, not just one?” Study I.f. transforms both variables x and y to be skewed (0.0, 0.5, 0.75, 1.0). As skewness rises, the correlations drop (0.298–0.263), t -statistics drop (5.46–4.78), and p -values rise (0.062–0.082). In the regressions with x as a predictor, the b -values drop (0.230–0.204) and the R^2 's drop (0.156–0.133). In the regressions with x as a dependent variable, the b -values drop (0.229–0.205), the R^2 's drop (0.158–0.133), and the standard errors rise (0.090–0.091), all conservative findings.

Study I.g. investigates a variant of the multivariate question, specifically, “What happens if two variables were skewed, but in opposite directions?” Variable x was transformed to be positively skewed and y to be negatively skewed (x at 0.0, 0.5, 0.75, 1.0 and y at 0.0, –0.5, –0.75, –1.0). These results are the first that are not dramatically conservative, rather, we refer to them as “somewhat conservative” in that the means for correlations drop or plateau to values that are roughly similar (0.298, 0.277, 0.281, 0.275), indicating that while no skewness is ideal (r is max at 0.298 with zero skewness), any level of skewness in x and y will yield a drop in the results. The t -statistics behave similarly (highest for no skew and drop as x and y are skewed

divergently (5.46–5.03, 5.14, and 5.01). Analogously p -values rise (0.061–0.075, 0.071, 0.073). In regressions with x as a dependent variable, the b -values drop (0.229–0.215), the R^2 's drop (0.158–0.135), and the standard errors rise (0.090–0.091).

In sum, Studies I.a.–I.g. offer a tentative conclusion that skewness does not matter much. Most effect sizes were very small (e.g., 1% variance accounted for, VAF), and even given the small effects, the results veered in the conservative direction. Specifically, as data became more skewed, b -coefficients tended to drop, R^2 's tended to drop, and standard errors tended to rise. That is, the studies indicated that skewed data would be less likely to yield significant results. While lack of significance can be disappointing to researchers, conservative statistics are in the preferred direction for scientific publishing and unambiguous scientific progress in hypothesis and theory testing.

4 | Study II: Kurtosis

Studies II.a.–II.g. examine the effect of kurtosis on the results of several frequently employed statistical models. Kurtosis is a function of the fourth moment of the normal distribution, X^4 , specifically $kurtosis = \frac{\sum(x_i - \bar{x})^4}{(n)\sigma^4}$ (Aron, Coups, and Aron 2012; Cribbie et al. 2012; Fabozzi et al. 2014; Ratcliff 1993; Sharma, Durvasula, and Dillon 1989; Wilcox 2021; for more detail, please see the Appendix). It reflects the extent to which the distribution appears peaked (leptokurtic, positive) or flat (platykurtic, negative) (DeCarlo 1997; see examples in the Appendix). In perfect standard normal distributions, kurtosis equals zero, and deviations are easily identified as statistical packages provide these indices with other descriptors, like means and standard deviations.

In these studies, standard normal distributions were generated for three variables (x, y, z) for sample sizes $N = 30, 100, 500, 1000$. Variable x was transformed to kurtosis levels of –1.0, –0.5, 0.0, 0.5, 1.0 (Fleishman 1978; and see DeCarlo 1997). Positive and negative kurtosis were both examined because these effects are not symmetric (positive kurtosis being peaked, negative being flat or boxy). For each sample size and kurtosis, we generated 1000 replications. For each of these $4 \times 5 \times 1000$ samples, we fit the same models that we had examined in Studies I.a.–I.f.

5 | Study II Results

Study II.a. examines three variables generated with modest intercorrelations ($\rho_{xy} = 0.3, \rho_{xz} = 0.3, \rho_{yz} = 0.3$). Table 2 shows that many but not all effects were significant. As with Studies I.a.–I.g., the significant findings are driven in part by power due to the size of the simulations (e.g., 1000 replications), given that the effect sizes are negligible, often explaining less than 1% of variance. For example, the effect of kurtosis on the t -statistics for correlations is significant, but the t -values vary little (5.35, 5.46, 5.47, 5.46, 5.46). The kurtosis levels were –1.0, –0.5, 0.0, 0.5, 1.0 so a statistical conservative result would show an inverted u-shape with smaller r 's, t 's, b 's, R^2 's at the left and right, and larger values in the center where kurtosis is zero, and standard errors would show a u-shape with the smallest in the center, bimodal growing larger to the left and right.

TABLE 2 | Results for kurtosis: *p*-values and effect sizes [Studies II.a.–II.g.].

	Study II.a.: r_{xy}'s 0.3	Study II.b.: r_{xy}'s 0.5	Study II.c.: r_{xy}'s 0.7	Study II.d.: r_{xy}'s 0.3, 0.5, 0.7	Study II.e.: greater kurtosis –1, 0, 1, 2	Study II.f.: X and Y multi- variate	Study II.g.: multivariate opposite
Correlations							
r_{xy}	n.s.	$p < 0.0001$ $\eta_p^2 = 0.002$	$p < 0.0001$ $\eta_p^2 = 0.006$	n.s.	$p < 0.0001$ $\eta_p^2 = 0.003$	$p < 0.0001$ $\eta_p^2 = 0.001$	$p = 0.0009$ $\eta_p^2 = 0.001$
t_{xy}	$p < 0.0001$ $\eta_p^2 = 0.002$	$p < 0.0001$ $\eta_p^2 = 0.007$	$p < 0.0001$ $\eta_p^2 = 0.027$	$p < 0.0001$ $\eta_p^2 = 0.002$	$p < 0.0001$ $\eta_p^2 = 0.013$	$p < 0.0001$ $\eta_p^2 = 0.005$	$p < 0.0001$ $\eta_p^2 = 0.003$
p_{xy}	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
Regression X predictor							
b_x	n.s.	$p < 0.0001$ $\eta_p^2 = 0.002$	$p < 0.0001$ $\eta_p^2 = 0.008$	n.s.	$p = 0.0106$ $\eta_p^2 = 0.001$	$p = 0.0007$ $\eta_p^2 = 0.001$	$p = 0.0091$ $\eta_p^2 = 0.001$
R^2	n.s.	n.s.	$p = 0.0069$ $\eta_p^2 = 0.001$	n.s.	$p = 0.0014$ $\eta_p^2 = 0.001$	$p = 0.0003$ $\eta_p^2 = 0.001$	$p = 0.0030$ $\eta_p^2 = 0.001$
std. error(x)	$p = 0.0065$ $\eta_p^2 = 0.001$	$p = 0.0251$ $\eta_p^2 = 0.001$	$p = 0.0028$ $\eta_p^2 = 0.001$	$p = 0.0019$ $\eta_p^2 = 0.001$	$p < 0.0001$ $\eta_p^2 = 0.003$	n.s.	$p < 0.0001$ $\eta_p^2 = 0.001$
Regression X dep var							
b_y	n.s.	$p = 0.0024$ $\eta_p^2 = 0.000$	$p < 0.0001$ $\eta_p^2 = 0.001$	n.s.	$p < 0.0001$ $\eta_p^2 = 0.005$	$p = 0.0002$ $\eta_p^2 = 0.001$	$p = 0.0077$ $\eta_p^2 = 0.001$
R^2	$p = 0.0062$ $\eta_p^2 = 0.001$	$p < 0.0001$ $\eta_p^2 = 0.002$	$p < 0.0001$ $\eta_p^2 = 0.010$	$p < 0.0001$ $\eta_p^2 = 0.001$	$p < 0.0001$ $\eta_p^2 = 0.004$	$p = 0.0014$ $\eta_p^2 = 0.001$	$p = 0.0133$ $\eta_p^2 = 0.001$
std. error(y)	n.s.	$p = 0.0084$ $\eta_p^2 = 0.001$	$p < 0.0001$ $\eta_p^2 = 0.003$	$p = 0.0266$ $\eta_p^2 = 0.001$	$p < 0.0001$ $\eta_p^2 = 0.003$	n.s.	$p < 0.0001$ $\eta_p^2 = 0.001$

Note: All results across kurtosis levels of –1, –0.5, 0, 0.5, and 1 were statistically conservative.

The regression in which x is a predictor shows significant results only for standard errors, with a small effect size (King and Roberts 2015). The means are rather flat or “uniform” (0.089, 0.90, 0.90, 0.91, 0.091). For the regression with x as a dependent variable, the only significant result is for R^2 's, and they do not particularly reflect an inverted-u to express statistically conservative effects of kurtosis (0.153, 0.157, 0.156, 0.154, 0.157). These findings are similar to the next kurtosis studies in being a mix of conservative findings and flat results.

Study II.b. examines larger correlations ($\rho_{xy} = 0.5, \rho_{xz} = 0.5, \rho_{yz} = 0.5$). Correlations were flat (0.489, 0.498, 0.497, 0.496, 0.496), and the t -statistics are slightly statistically conservative (inverted u, 9.77, 10.00, 10.02, 10.02, 10.00). In the regressions with x as a predictor, the b -values and standard errors are rather flat (b 's 0.321, 0.333, 0.332, 0.333, 0.333, standard errors from 0.087 to 0.088). In the regressions with x as a dependent variable, the results on b 's, R^2 's, and standard errors are all somewhat flat (b 's 0.325, 0.333, 0.331, 0.333, 0.334, R^2 's 0.335, 0.345, 0.344, 0.341, 0.344, S.E.'s 0.088 to 0.087). We elaborate further on the combination of conservative and flat results after we present the remaining Studies II.c.–II.g.

Study II.c. increases the intercorrelations ($\rho_{xy} = 0.7, \rho_{xz} = 0.7, \rho_{yz} = 0.7$). The correlations and t -statistics are conservative (inverted u, r 's = 0.685, 0.697, 0.697, 0.696, 0.696, t 's = 16.4, 16.96, 17.0, 16.97, 16.94). In the regressions with x as a predictor, the b -values and the R^2 's drop a bit at the ends and appear somewhat conservative or flat (b 's 0.388 to 0.410, R^2 's 0.576, 0.576, 0.581, 0.580, 0.580). The standard errors look flat (0.083–0.085). In the regressions with x as a dependent variable, the b -values and standard errors are flat (b 's 0.402, 0.411, 0.410, 0.412, 0.409, S.E.'s 0.087–0.084), and the R^2 's look somewhat conservative being a bit larger in the center when kurtosis is zero (i.e., 0.562, 0.581, 0.581, 0.578, 0.578).

Study II.d. considers a mix of intercorrelations ($\rho_{xy} = 0.3, \rho_{xz} = 0.5, \rho_{yz} = 0.7$). Table 2 shows few significant results, and again, very small effect sizes. The t -statistics for the correlations are somewhat conservative (5.35, 5.46, 5.47, 5.46, 5.46), and the standard errors in both regressions are flat (0.075–0.077 and 0.113–0.111), as are the R^2 's in the regressions with x as a dependent variable (i.e., 0.113, 0.112, 0.112, 0.112, 0.111).

Study II.e. extends the kurtosis. Fleishman (1978, 526) notes that a perfectly rectangular distribution is associated with the

minimal value of kurtosis being -1.2 . We had investigated effects of $-1.0, -0.5, 0.0, 0.5, 1.0$, thus there was little room to extend the platykurtic (negative) side. However, the positive (peaked) kurtosis could be increased. In Study II.e., we extended kurtosis to range from $-1, 0, 1, 2$. Table 2 shows that greater kurtosis leads nearly uniformly to more statistically conservative findings. The correlations were slightly larger for the value of zero kurtosis (the second value in the listing: 0.293, 0.298, 0.297, 0.0296, similarly for t 's 5.35, 5.47, 5.46, 5.43). For the regressions, the b 's were somewhat conservative (0.225, 0.230, 0.230, 0.229 and 0.225, 0.230, 0.230, 0.228), and the R^2 and standard error results were conservative and flat (R^2 's of 0.154, 0.156, 0.156, 0.155 and 0.153, 0.156, 0.157, and 0.155, and standard errors were 0.089–0.090 and 0.091–0.089).

Study II.f. transforms both variables x and y to show varying levels of kurtosis. Most results are statistically conservative. The r 's and t -statistics yield maximum values at the center when kurtosis is zero (r 's 0.288, 0.298, 0.298, 0.297, 0.296, t 's 5.26, 5.44, 5.47, 5.46, 5.45). In regressions with x as a predictor, the b 's and R^2 's peak a bit in the center but are rather flat (b 's 0.221, 0.230, 0.230, 0.230, 0.229, R^2 's 0.152, 0.157, 0.157, 0.154, 0.157). In regressions with x as a dependent variable, the b 's peak a bit (0.221, 0.230, 0.230, 0.231, 0.229), and R^2 's appear flat (0.152, 0.157, 0.157, 0.154, 0.157).

Study II.g. transforms x and y to show inverse kurtosis (x took values $-1.0, -0.5, 0.0, 0.5, 1.0$ while y took values $1.0, 0.5, 0.0, -0.5, -1.0$). Statistically conservative results for r , t , b , or R^2 would be larger in the center where neither x and y exhibit kurtosis. Study II.g. shows a mix of results that are statistically conservative findings with peaking (e.g., r 's 0.291, 0.298, 0.298, 0.296, 0.291, t 's 5.33, 5.45, 5.47, 5.45, 5.34, R^2 0.153, 0.155, 0.157, 0.156, 0.151 and flat 0.153, 0.157, 0.157, 0.153, 0.155) and standard errors appear flat in both regressions (0.089–0.091, and 0.091–0.090).

In summary, the kurtosis Studies II.a.–II.g. were not as simple as the skewness Studies I.a.–I.g. in that these were a mix of classically statistically conservative findings (e.g., b 's declining) and somewhat conservative or flat findings. Nevertheless, that combination, coupled with the tiny effect sizes, might still suggest that scholars need not worry too much about variations of kurtosis.

6 | Study III: Outliers

Studies III.a.–III.g. examines outliers. Whereas skewness and kurtosis and deviations thereof are defined precisely as a function of statistical theory, outliers are defined more as a function of scientific consensus, as “an observation (or subset of observations) which appears to be inconsistent with the remainder of that set of data” (Barnett and Lewis 1994, 4; Pollet and van der Meij 2017). In particular, there is some consensus regarding declaring outliers to be data points with magnitudes beyond 2.5 or 3.0 standard deviations from the mean (Aguinis, Gottfredson, and Joo 2013; Analytical Methods Committee [AMC] 1989; Andersen 2008; Barnett and Lewis 1994; Belsley, Kuh, and Welsch 1980; Boos and Stefanski 2015; Cook and Weisberg 1982; Daszykowski et al. 2007; Dixon 1953; Grubbs 1969; Huber and Ronchetti 2009; Leone, Minutti-Meza, and Wasley 2019;

Maronna et al. 2018; Osborne and Overbay 2004; Stefansky 1972; Sullivan, Warkentin, and Wallace 2021; Tukey 1962; Y. Yuan and MacKinnon 2014). The standard deviation cut-off rule is frequently employed in journal articles, but other techniques for identifying outliers have been offered, such as the use of boxplots (Tukey 1977) which can be helpful given the problem of “masking,” whereby the deletion of an extreme datapoint can change the data set's standard deviation, in turn making the cut-off elusive (Wilcox and Keselman 2003; Wilcox 2017; see the Appendix for more details).

Accordingly, in Studies III.a.–III.g, we created the outliers by adding the value 3.0 to the relevant percentage (e.g., 5%) of the positive values selected randomly in the standard normal distribution of x . Standard normal distributions were generated for three variables (x, y, z) for sample sizes $N = 30, 100, 500, 1000$. The percentage of outliers on variable x included 0%, 1%, 5%, 10%. We also explored larger percentages (25% and 33%), but these percentages were so great that the distributions became bimodal, which manifests altogether different statistical issues, beyond the scope of this paper. In addition, while the deletion of up to 10% of one's data may raise doubts in the minds of reviewers and readers, the deletion of 25% or 33% would surely signal a problematic study and possibly dubious scholarly actions. For each sample size and percentage of outliers, we generated 1000 replications, and for each of the $4 \times 4 \times 1000$ samples, we fit the models reported in Studies I and II.

7 | Study III Results

Study III.a. examines modest intercorrelations ($\rho_{xy} = 0.3, \rho_{xz} = 0.3, \rho_{yz} = 0.3$). In Table 3, for outlier levels of 0%, 1%, 5%, 10%, statistically conservative r 's, t 's, b 's, and R^2 's would decrease from 0% to 10% as the percentage of outliers rose. Conservative results for p -values and standard errors would increase. Indeed, the correlations trend downward (0.298, 0.291, 0.260, 0.247), their t -values downward (5.46, 5.27, 4.74, 4.50), and the p -values upward (0.060, 0.63, 0.081, 0.089). The regression with x , the variable with outliers as a predictor showed standard errors that declined (0.090–0.057), but they decreased slower than the b -values (0.230–0.120), hence the t -statistics show conservatism (2.56, 2.44, 2.15, 2.11). Similarly, for regressions in which x served as a dependent variable with outliers, the b -values rose, but not as quickly as the standard errors, hence, again, the t -statistics showed conservatism (2.56, 2.53, 2.22, 2.07).

Study III.b. examines slightly higher correlations ($\rho_{xy} = 0.5, \rho_{xz} = 0.5, \rho_{yz} = 0.5$). The correlations fall (0.498, 0.484, 0.435, 0.413), as do the t -values (10.01, 9.56, 8.44, 7.94), and the p -values rise (0.009, 0.009, 0.020, 0.026). In the regressions with x as a predictor variable, the standard errors decreased (0.088–0.054), as did the b -values (0.332–0.162), and the t -statistics fell (from 3.77 to 3.00). Similarly, in the regressions with x as the dependent variable, the b -values rose (0.331–0.431), as did the standard errors (0.087–0.146), and the t -statistics fell conservatively (3.80–2.95).

Study III.c. increases the intercorrelations ($\rho_{xy} = 0.7, \rho_{xz} = 0.7, \rho_{yz} = 0.7$). Correlations dropped (0.698, 0.678, 0.611,

TABLE 3 | Results for outliers: p -values and effect sizes [Studies III.a.–III.g.].

	Study III.a.: r_{xy}'s 0.3	Study III.b.: r_{xy}'s 0.5	Study III.c.: r_{xy}'s 0.7	Study III.d.: r_{xy}'s 0.3, 0.5, 0.7	Study III.e.: greater outliers	Study III.f.: X and Y multivariate	Study III.g.: multivariate opposite
Correlations							
r_{xy}	$p < 0.0001$ $\eta_p^2 = 0.041$	$p < 0.0001$ $\eta_p^2 = 0.136$	$p < 0.0001$ $\eta_p^2 = 0.363$	$p < 0.0001$ $\eta_p^2 = 0.041$	$p < 0.0001$ $\eta_p^2 = 0.219$	$p < 0.0001$ $\eta_p^2 = 0.828$	$p < 0.0001$ $\eta_p^2 = 0.926$
t_{xy}	$p < 0.0001$ $\eta_p^2 = 0.122$	$p < 0.0001$ $\eta_p^2 = 0.346$	$p < 0.0001$ $\eta_p^2 = 0.658$	$p < 0.0001$ $\eta_p^2 = 0.122$	$p < 0.0001$ $\eta_p^2 = 0.495$	$p < 0.0001$ $\eta_p^2 = 0.956$	$p < 0.0001$ $\eta_p^2 = 0.892$
p_{xy}	$p < 0.0001$ $\eta_p^2 = 0.006$	$p < 0.0001$ $\eta_p^2 = 0.010$	$p < 0.0001$ $\eta_p^2 = 0.009$	$p < 0.0001$ $\eta_p^2 = 0.006$	$p < 0.0001$ $\eta_p^2 = 0.049$	$p < 0.0001$ $\eta_p^2 = 0.079$	$p < 0.0001$ $\eta_p^2 = 0.117$
Regression X predictor							
b_x	$p < 0.0001$ $\eta_p^2 = 0.174$	$p < 0.0001$ $\eta_p^2 = 0.350$	$p < 0.0001$ $\eta_p^2 = 0.543$	$p < 0.0001$ $\eta_p^2 = 0.032$	$p < 0.0001$ $\eta_p^2 = 0.113$	$p < 0.0001$ $\eta_p^2 = 0.812$	$p < 0.0001$ $\eta_p^2 = 0.909$
R^2	$p < 0.0001$ $\eta_p^2 = 0.009$	$p < 0.0001$ $\eta_p^2 = 0.024$	$p < 0.0001$ $\eta_p^2 = 0.052$	n.s.	n.s.	$p < 0.0001$ $\eta_p^2 = 0.844$	$p < 0.0001$ $\eta_p^2 = 0.622$
std. error (x)	$p < 0.0001$ $\eta_p^2 = 0.423$	$p < 0.0001$ $\eta_p^2 = 0.440$	$p < 0.0001$ $\eta_p^2 = 0.485$	$p < 0.0001$ $\eta_p^2 = 0.456$	$p < 0.0001$ $\eta_p^2 = 0.803$	$p < 0.0001$ $\eta_p^2 = 0.333$	$p < 0.0001$ $\eta_p^2 = 0.121$
Regression X dep var							
b_y	$p < 0.0001$ $\eta_p^2 = 0.037$	$p < 0.0001$ $\eta_p^2 = 0.078$	$p < 0.0001$ $\eta_p^2 = 0.106$	$p < 0.0001$ $\eta_p^2 = 0.003$	$p < 0.0001$ $\eta_p^2 = 0.009$	$p < 0.0001$ $\eta_p^2 = 0.821$	$p < 0.0001$ $\eta_p^2 = 0.910$
R^2	$p < 0.0001$ $\eta_p^2 = 0.068$	$p < 0.0001$ $\eta_p^2 = 0.215$	$p < 0.0001$ $\eta_p^2 = 0.488$	$p < 0.0001$ $\eta_p^2 = 0.152$	$p < 0.0001$ $\eta_p^2 = 0.510$	$p < 0.0001$ $\eta_p^2 = 0.840$	$p < 0.0001$ $\eta_p^2 = 0.651$
std. error (y)	$p < 0.0001$ $\eta_p^2 = 0.501$	$p < 0.0001$ $\eta_p^2 = 0.545$	$p < 0.0001$ $\eta_p^2 = 0.636$	$p < 0.0001$ $\eta_p^2 = 0.526$	$p < 0.0001$ $\eta_p^2 = 0.888$	$p < 0.0001$ $\eta_p^2 = 0.301$	$p < 0.0001$ $\eta_p^2 = 0.132$

Note: All results in Studies III.a.–III.e. across outlier percentages of 0%, 1%, 5%, and 10% were statistically conservative. Results for multivariate Studies III.f. and III.g. were not conservative.

0.579), as did t -statistics (17.01, 15.89, 13.43, 12.42), and p -values rose (0.0002–0.003). In the regressions with x as a predictor, the standard errors drop (0.085–0.049), but the b -values drop faster (0.411–0.170), so the t 's are conservative (from 4.84 to 3.47). In the regressions with x as a dependent variable, the b -values increase (0.409–0.532) but the standard errors increase faster (0.084–0.157), so the resulting t 's are conservative (4.87–3.39).

Study III.d. considers a mix of correlations ($\rho_{xy} = 0.3, \rho_{xz} = 0.5, \rho_{yz} = 0.7$). The correlations fall (0.298–0.247), as do the t -statistics (5.46–4.50), and the p -values rise (0.061–0.089). In the regressions with x as a predictor, the standard errors decrease (0.076 to 0.046), the b -value magnitudes drop faster, so the t 's are conservative in magnitude (–0.91 to –0.70). In the regressions with x as a dependent variable, the b -values increase and the standard errors rise faster (0.112–0.184), so the resulting t -value magnitudes are conservative (–0.91 to 0.67).

Before proceeding to Studies III.e.–III.g., it may be of interest to note that the effect sizes in Table 3 are larger than the tiny effect

sizes seen thus far in Tables 1–2 (those often being 0.001). In some cases, these η_p^2 are much larger. Such results would be cause for concern if it had not continued to be the case that the results appear to be flat or statistically conservative.

Study III.e. extends the outliers. Thus far, we have created outliers by generating data points beyond 3 standard deviations, yet perhaps it is not the quantity of outliers that might affect data analyses as much as the magnitude of the outliers. In Study III.e., we looked at the same percentage of outliers, 0%, 1%, 5%, and 10%, but in the creation of the outliers, instead of adding the value 3 (for 3 standard deviations), we added the value of 10. Even with this substantial boost in the magnitude of the problematic data points, the statistically conservative findings show, correlations drop (0.298, 0.242, 0.169, 0.168), t -values drop (5.46, 4.10, 3.08, 3.04), and p -values rise (0.062–0.148). In regressions with x as a predictor, standard errors drop (0.076 to 0.019), and the b -values drop faster, so the t -values are conservative (–0.91 to –0.42). In regressions with x as a dependent variable, the b -values increase in magnitude (–0.102 to –0.183), the standard errors also rise (0.112–0.436), and t -value magnitudes are

conservative (-0.91 , -0.76 , -0.49 , -0.42). The R^2 also drop (0.271 , 0.187 , 0.104 , 0.100).

The studies in III examine whether the presence of an outlier on one variable may affect results pertaining to the empirical correlations with other, normally distributed variables. Study III.e. in particular focuses on larger magnitudes of outliers compared with Studies III.a.–III.d. One side of the coin is to ask whether a variable with outliers can affect the results regarding relationships with other (normally distributed) variables. It seems likely; imagine a three-dimensional scatterplot—much of it might look like a trivariate normal distribution, but one variable is pulling the plot in a more extreme direction. Results in Table 3 for Study III.e. indeed indicate significance and non-zero albeit modest effect sizes for the parameters (r 's, b_x 's, b_y 's). The other side of the coin is to ask whether the normally-distributed variables have an impact on the results reflecting the relationships including the variable with outlier, might they dampen down and reduce the distortion of a variable with outliers to resemble a more normal-looking trivariate distribution. Table 3 for Study III.e. indicates that the effect size for the standard errors are up (for the two regression panels of results), indeed by quite a lot, very likely attributable to the resulting abnormalities of the data—two normal variables, one not. Together, the results lend to our overall statements reflecting the impact veering toward the conservative.

Study III.f. creates outliers on both variables x and y . For this study, *none* of the results are statistically conservative. For correlations, r 's rose from 0.298 , 0.371 , 0.595 , and 0.737 , as did the t -values (5.46 – 18.86), and p -values fell (0.062 – 0.000). In regressions with x as the predictor, b 's were 0.230 , 0.314 , 0.566 , 0.714 , and the t -values are not conservative (2.56 – 11.33), and similarly for the regressions with x as the dependent variable (2.54 – 11.33). These findings are obviously problematic. We will address these patterns after describing the results for Study III.g. For now, we note that the presence of outliers on two variables explodes the statistical results in a liberal fashion (more likely to find significance) and do so in a dramatic fashion (note the large effect sizes).

Study III.g. creates outliers on both variables x and y in opposite directions (x took outlier values of 0% , 1% , 5% , and 10% while y took values of 10% , 5% , 1% , 0%). Mean values at the center should be somewhat larger for r , t , b , or R^2 if the results are statistically conservative. The results are mixed. Correlations were 0.298 , 0.166 , -0.253 , -0.514 , t 's were 5.46 , 2.45 , 4.94 , 10.32 , though p -values were conservative, peaking in the center, 0.062 , 0.146 , 0.108 , 0.004 . The two regressions showed similarly mixed results: b 's (0.230 , 0.095 , -0.321 , -0.566 and 0.229 , 0.094 , -0.310 , and -0.552), R^2 (0.156 , 0.108 , 0.149 , 0.330 and 0.158 , 0.120 , 0.181 , 0.365), and standard errors (0.090 , 0.092 , 0.090 , 0.077 and 0.090 , 0.091 , 0.087 , 0.075).

Together, Studies III.f. and III.g. indicate that multivariate outliers are problematic and may possibly lead researchers to misleading conclusions. In retrospect, the results of Studies III.f. and III.g. are perhaps not surprising. As taught in introductory statistics classes, a single outlying data point has the potential to change the slope of a regression line (Credé 2010). How much more dramatic might this effect be if there were 1% , 5% , or 10%

of outliers increasing some values of x and y (per Study III.f.) or increasing x and decreasing y (per Study III.g.)?

In summary, the first five outlier Studies III.a.–III.e. were consistent with the findings on skewness and kurtosis in demonstrating that the particular class of nonnormal data (skewness, kurtosis, outliers) would largely lead to results that were more statistically conservative (less likely to yield significant results). Thus, like skewness and kurtosis, scholars need not be overly concerned with univariate outliers. Studies III.f. and III.g. were different, identifying a boundary condition that multivariate outliers can be a problem. Next, we explain how to proceed.

8 | Discussion

Overall, the results of Studies I, II, and III.a.–III.e. were rather straightforward and encouraging in that scholars need not worry too much about nonnormal and potentially problematic data. Skewness, kurtosis, and univariate outliers typically yielded results that were very small in effect size, and statistically conservative in direction. Studies III.f. and III.g. however, increased the complexity of the story, which we return to shortly. Next, we discuss the contributions of our research and provide guidance to scholars.

In Studies I.a.–I.g., the findings seem to indicate that researchers should not be overly concerned with skewed variables. When skewness modified analytical outcomes, they did so with minute effect sizes, and the results veered in a conservative direction. The presence of skewness seems to lead to outcomes which are less likely to be deemed significant, the direction that supports journals and scientific progress. If the skewed data had trended in the opposite direction, readers and reviewers could worry that the results in a paper had been inflated, reporting more significant findings than had the data been cleaner. Furthermore, the field would have had greater challenges with replications and meta-analyses. However, skewness is a data characteristic that does not seem to cause any significant problems.

Studies II.a.–II.g. considered kurtosis in an analogous fashion and the results were slightly more complicated, but the conclusions were no more concerning. Analytical results primarily showed the conservative trend documented for skewness. In addition, many results exhibited what we called a softer conservative finding of flatness, in that the results for any level of kurtosis either dropped in a conservative direction or simply appeared unaffected. Thus, scholars encountering a variable that is flat or peaked may not be excessively worried in that their results are less likely to produce significance compared to clean data, that is, data that show zero kurtosis.

Studies III.a.–III.e. on outliers suggested that such data are not a critical concern—the worse that might happen is that the data would not yield significant findings. However, Studies III.f. and III.g. provided different results. Specifically, those studies examined the multivariate problem of two variables with outliers, in the same or opposite directions, and results tracked toward significance. Interestingly, we believe that the latter two studies

on outliers help support the importance of this research—namely, when do nonnormal data matter? Studies III.f. and III.g. offer boundary conditions that clearly explicate conditions when scholars need to be concerned with nonnormal data—skewness, kurtosis, or outliers. If one's data show skewness, kurtosis, or even univariate outliers, the results across the studies show rather consistently that nonnormal data typically led to fewer significant results than normal data (and statistical conservatism is a valued scientific quality). When scholars check their data in initial cleaning and obtaining preliminary descriptive statistics, they can identify variables with outliers, and not be too worried if some variable is the only variable in a particular model with outliers. Yet, if, for example, a regression is to be fit wherein more than one variable is affected by outliers, researchers need to engage in data cleaning exercises that would not be necessary for the presence of a single variable with univariate outliers.

Our research contributes to the literature by examining more versions of aberrant data and more models than have been studied to date (e.g., Barbato et al. 2011; Franses, Kloek, and Lucas 1998; Pollet and van der Meij 2017; Rice and Spiegelhalter 2006). The focus of our research was to examine how results in data analysis would be affected if such data were included. In our research, we sought to understand *when or whether* researchers might need to worry about such data. This is particularly relevant in today's ecosystem, where there have been scandals and accusations concerning data manipulation. Our results indicate that researchers do not need to delete or otherwise manipulate data that are a function of skewness, kurtosis, or univariate outliers. These results are also welcome because they help avoid potential reviewer critiques that the data were somehow inappropriately manipulated.

Any research has limitations that may be pursued in future research. For example, we selected three primary ways that data may look nonnormal or potentially problematic, yet scholars working in different research domains may more often encounter data with different properties, such as bimodal data, perhaps reflecting a mixture of distinct latent segment distributions. Such an occurrence may well cause analytical challenges. Similarly, we focused on data that deviated from normality assumptions as pertaining to many frequently used statistical analyses, but for researchers who more often implement analyses that assume gamma, Poisson, or other distributions, those violations could also be studied. Further, while we examined sample sizes that we thought were reasonably representative, ranging from 30 to 1000, researchers who routinely work with small numbers of case studies (say $N = 4$ or 5 in some medical settings) may wish to study the impact of nonnormalities in such small samples. Conversely, researchers who work with “big data” may wish to examine larger samples. Finally, in some research domains, different statistical models may be more prevalent than those we studied here, so research could be extended to those models as well.

To further support the findings from our simulations, we performed additional analyses on six real-world data sets (cf., Supporting Information S1: [Web Appendix](#)). The main insight from analyzing those data sets is that the conclusions of our paper are well-supported. Researchers should not be overly

concerned about nonnormalities in their data, as our findings also hold true across different real-world contexts. Yet, if more than one variable is affected by outliers in a regression model, researchers would need to clean their data.

While this paper uses best practices in statistics and simulations, the analyses in the Supporting Information S1: [Web Appendix](#) offer additional, complementary evidence from real-world data sets, confirming the reliability of the results.

9 | Conclusion

Presumably the major philosophical scientific concern about nonnormal data is that certain statistics might be inflated toward a greater likelihood of finding significance. Journal norms usually operate under conservative philosophies, that is, the bar is high for demonstrating an effect, and if the result surpasses the bar, the significant results should be published. No one wants the fruit of their research labors to be a function of ill-behaved data. If research were to be published that did not meet a conservative bar, there may be less likelihood of the replicability of the purported findings.

A priori, we had anticipated that significance tests and parameter estimation bias would be dramatic in the face of skewness, kurtosis, and outliers. Yet empirically, our studies showed the analyses were typically dampened toward conservative findings, that is, nonsignificance. Two studies showed exceptions—when data exhibit multivariate outliers, results can be problematic. Thus, let us consider what these results portend for researchers. We are not trying to create new “rules” but given our findings, we are simply offering suggestions of changes that could help researchers.

1. We would suggest that scholars need not, indeed perhaps should not, delete data if they are concerned about skewness, kurtosis, or univariate outliers. If there are nonnormal data, the findings are simply likely to veer toward being insignificant. Insignificant findings can be disappointing to the research team, but if data are deleted to make them look less skewed (for example), questions may arise as to whether that data processing step was necessary or done to enhance the significance of findings.
2. The exception to the first guideline is when scholars detect outliers on at least two variables that are entered into a focal model. When this pattern of nonnormal data occurs, researchers will return to the status quo of deleting or transforming the multivariate outlier values.
3. If, per the second guideline, researchers wished to delete or transform data, they should provide precise documentation in Supporting Information S1: [Web Appendices](#) regarding the changes (cf., Aguinis, Gottfredson, and Joo 2013; Beaumont and Rivest 2009; Dixon 1953; R. J. Lawrence 1980). The data sets with and without the deleted data points should be posted, and the deleted data points highlighted in the full data set to make the actions clear. Similarly, scholars should post results on both data sets, detailing any differences. Full documentation would be important for transformations or alternative methods used

as well. For example, when techniques such as Winsorizing are used, the decisions (e.g., the number of observations to Winsorize) should be documented and justified (e.g., “Why three, why not five?” etc.). Doing so will be challenging, as there is little guidance regarding proper norms.

Overall, our research indicated that the impact of nonnormal data—in the form of skewness, kurtosis, or univariate outliers—were typically negligible and nudged the results in the statistically conservative directions toward nonsignificance. An advantage of retaining all of one's data and knowing that their results will be sound is that with such full data usage, researchers do not have to worry about explaining any data treatment. Data with multivariate outliers, on the other hand, are problematic and scholars need to document any data adjustments they invoke.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Appendix A

Brief Tutorial Refresher to Skewness, Kurtosis, and Outliers

Concept Definitions and Theoretical Foundations

In this tutorial refresher, we: (1) define the concepts of skewness, kurtosis, and outliers, (2) show how they are identified in data, and (3) discuss several solutions that statisticians have offered in the past.

Definitions

To understand three important forms of nonnormal data, we begin with the parameters for skewness and kurtosis, and then define outliers. As presented in statistics texts (e.g., Aron, Coups, and Aron 2012), in a standard normal distribution, the statistical parameters that characterize its location and shape are functions of the “moments” of the distribution (X , X^2 , X^3 , and X^4). The sample mean is a function of X to reflect location ($\bar{x} = \frac{\sum x_i}{n}$), the sample variance a function of X^2 to reflect dispersion ($\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$) or heteroscedasticity, skewness is a function of X^3 ($skew = \frac{\sum (x_i - \bar{x})^3}{(n)\sigma^3}$) to reflect symmetry versus asymmetry, and kurtosis of X^4 ($kurtosis = \frac{\sum (x_i - \bar{x})^4}{(n)\sigma^4}$) to reflect flatness versus peakedness. (Unbiased formulae for skewness and kurtosis use multipliers $\frac{n}{(n-1)(n-2)}$ and $\frac{(n+1)n}{(n-1)(n-2)(n-3)}$ rather than $\frac{1}{n}$.)

Skewness, recall, is positive if there are some larger than normal observations (including outliers) in the right tail (and the mean is greater than the median), and negative if such observations appear in the left tail (see Figure A1). Kurtosis reflects the extent to which the distribution is peaked (leptokurtic, positive) or flat (platykurtic, negative; DeCarlo 1997), as depicted in Figure A2.

In a perfect standard normal (z) distribution, skewness and kurtosis values equal 0. To provide a sense of the range of these values,

Figure A3 shows examples of distributions generated that vary in skewness and kurtosis. Each sample represents $N = 1000$.

In contrast, outliers are not defined as parameters or moments of a distribution. Instead, outliers are defined less formally as “an observation (or subset of observations) which appears to be inconsistent with the remainder of that set of data” (Barnett and Lewis 1994, 4). Pollet and van der Meij (2017) similarly define outliers as those data points that appear to be different from most other observations in the sample. More precisely, outliers are typically defined as data points with magnitudes beyond 2.5 or 3.0 standard deviations from mean (see Aguinis, Gottfredson, and Joo 2013; Daszykowski et al. 2007; Grubbs 1969; Stefansky 1972). Figure A4 illustrates outliers on the right side of the normal curve.

Detection

There is no ambiguity in detecting skewness or kurtosis. Standard statistical software (e.g., SAS, SPSS) produce these parameters as an option along with the provision of other descriptive statistics such as means and standard deviations. Normal distributions would have scores for skewness and kurtosis at zero or close to zero, and other values for skewness and kurtosis can be compared to the samples illustrated in Figure A3 to understand the nature of the distribution's shape.

While outliers are not a distribution parameter and could be characterized as somewhat more subjective, the detection of outliers has converged in the literature to primarily two approaches. The first, as mentioned, is to declare a data point as an outlier if it equals or exceeds 2.5 (or 3.0) standard deviations (i.e., $z \geq 2.5$; Daszykowski et al. 2007; Pollet and van der Meij 2017; Sullivan, Warkentin, and Wallace 2021; and see Aguinis, Gottfredson, and Joo 2013 for more outlier identification techniques). Rice and Spiegelhalter (2006) describe the logic, that for a normal distribution, theoretically 95% of the data should lie between -2.0 and 2.0 standard deviations. While, of course, values can exceed this range—theoretically to negative and positive infinity—those extreme values are, by definition, unlikely. Hence a cutoff beyond $|2.0|$, or $|2.5|$ to be safer, is deemed noncontroversial.

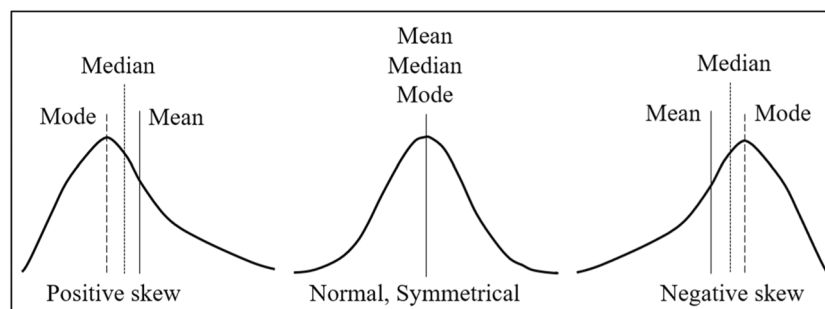


FIGURE A1 | Skewness.

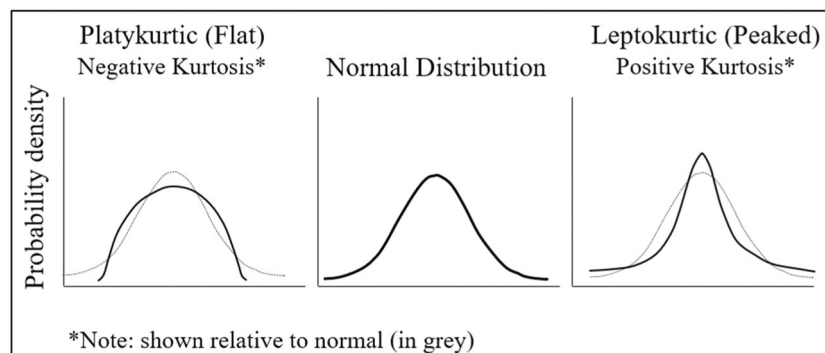
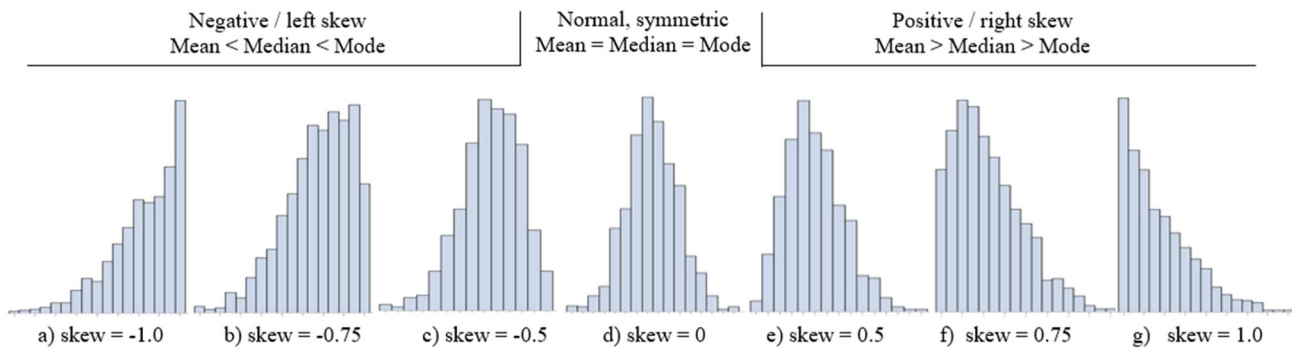


FIGURE A2 | Kurtosis.

(a) Skewness:



(b) Kurtosis:

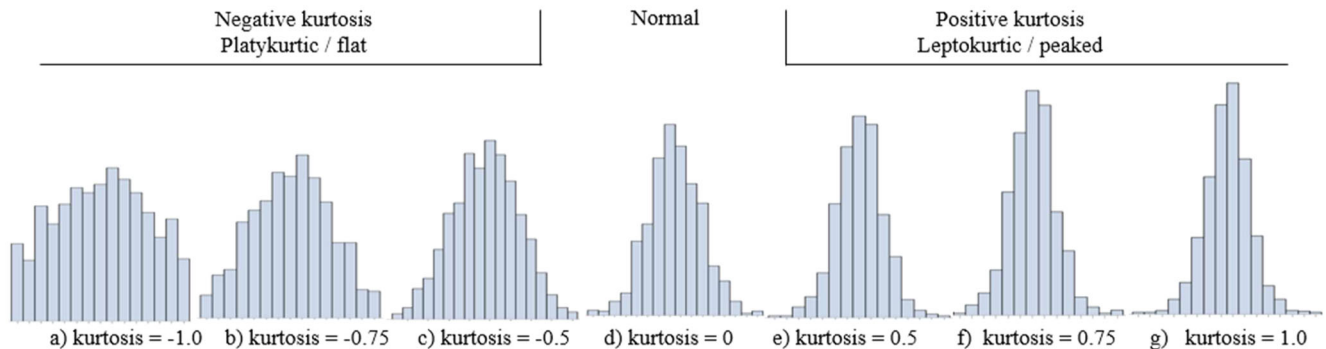


FIGURE A3 | Degrees of skewness and kurtosis.

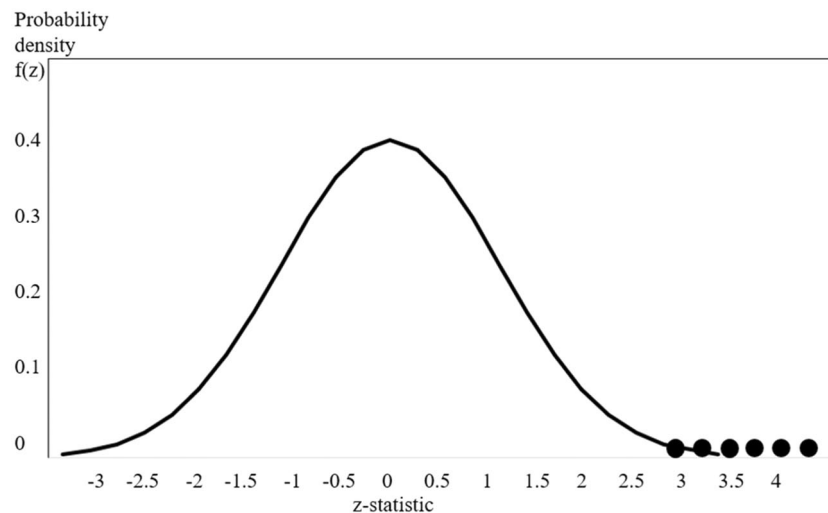
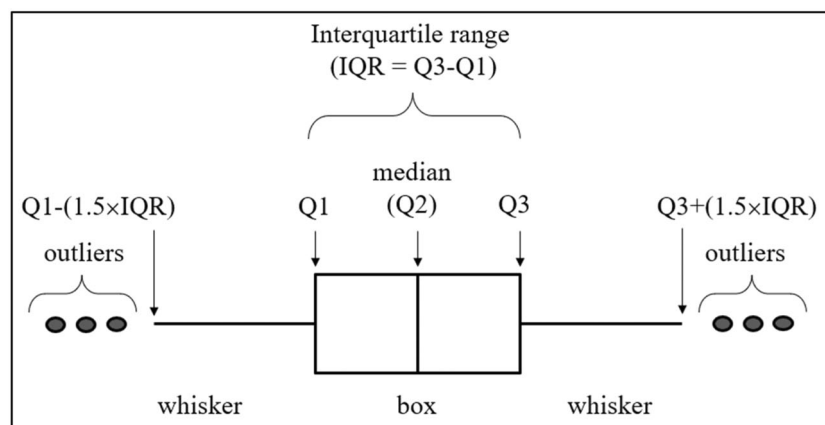


FIGURE A4 | Outliers on the right-hand side of the normal curve.

The second approach to detecting and defensibly declaring a data point to be an outlier is based on boxplots (Tukey 1977). Plots are helpful in part due to the problem of “masking” whereby the determination of a z -score of 2.5 is based on a standard deviation that would change with the extraction of an extreme data point, making the cutoffs slightly elusive (Wilcox and Keselman 2003; Wilcox 2017). Given the masking problem, boxplots are the superior means of identifying outliers, but methods sections in journal articles are dominated by the use of heuristics of z -values exceeding 2.5, and boxplots are rarely used. The algorithm for using a boxplot to identify outliers is simple (explained next) and easily accessible, produced by both SAS and SPSS (Barbato et al. 2011; Daszykowski et al. 2007; Rice and Spiegelhalter 2006; Wilcox and Keselman 2003).

Figure A5 depicts a boxplot. The box is built with its left (or bottom) side at the value of the variable's first quartile, Q_1 (the 25th percentile of standard normal, $z = -0.675$) and its right side at the third quartile, Q_3 (the 75th percentile, $z = 0.675$). The width of the box is the difference called the interquartile range (IQR) = $Q_3 - Q_1$ (i.e., $0.675 - (-0.675) = 1.35$). IQR essentially removes more extreme observations which may be outliers. Lines called “whiskers” are added that have lengths 1.5 times IQR (i.e., $(1.5)(1.35) = 2.025$) to extend the box to the left and right. Whiskers thus form a minimum of $-0.675 - 2.025 = -2.7$ at the low end, and $0.675 + 2.025 = 2.7$ at the high end. Outliers are said to be any data points that fall beyond the whiskers. Note that these values, -2.7 and 2.7 correspond roughly to the 2.5 or 3 standard deviations as suggested per the literature.



Note: $Q1 = 1^{\text{st}}$ quartile = 25th percentile; $Q3 = 3^{\text{rd}}$ quartile = 75th percentile.

FIGURE A5 | Box plot for determining outliers. Note: $Q1 = 1^{\text{st}}$ quartile = 25th percentile; $Q3 = 3^{\text{rd}}$ quartile = 75th percentile.

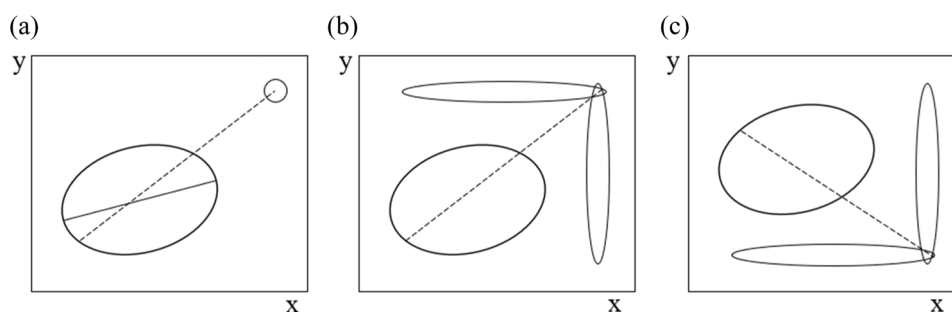


FIGURE A6 | Outliers influencing empirical relationships. (a) Depicts the pull of an outlier on a correlation scatter plot; (b and c) indicate similar effect of outliers that are large for x and y (A6.b.) or large for x and small for y (A6.c.).

In general, outliers can seem problematic, by themselves, or when many large values are concentrated in one of the tails, contributing to skewness, or in both (or neither) tails, reflected in kurtosis (Wilcox and Keselman 2003). Figure A6 shows how outliers can affect results such as slopes of lines in regressions.

In the literature, numerous solutions to nonnormal data have been offered. Most are rarely implemented, thus perhaps unfamiliar to many scholars, so next we highlight several of the standard recommendations.

Current Solutions from the Robustness Literature

Statisticians have long raised the concern that nonnormal data might *theoretically* affect analytical results (Box and Watson 1962; Hogg 1979; Huber and Ronchetti 2009; Stahel and Weisberg 1991). This concern seems logical theoretically, and the statistical literature emphasizes the derivations of robust statistics as solutions (discussed next, e.g., K. D. Lawrence 2020). The robustness literature typically focuses on one form of nonnormality (e.g., skewness or kurtosis or outliers) and one particular model, often to make a case for using a newly proposed statistical index (e.g., Belsley, Kuh, and Welsch 1980; Maronna et al. 2018; Wilcox 2021).

Our research addresses whether analytical results are in fact adversely affected by nonnormal data. That is, our research starts more fundamentally, to offer a novel and important contribution in that we examine whether the theoretical predictions of adversely affected results are actually borne out *empirically*. We do not seek to promote another new index or technique. Instead, we examine variants of nonnormalities and models to develop an enhanced and more comprehensive picture of the

impact of nonnormal data, and better benchmarks to draw richer conclusions.

The statistical literature on robustness offers two major classes of solutions to address nonnormal data (Andersen 2008; Bakker and Wicherts 2014; Cook and Weisberg 1982; Daszykowski et al. 2007; Fabozzi et al. 2014; Hettmansperger and McKean 2010; Leone, Minutti-Meza, and Wasley 2019; Maronna et al. 2018; Pollet and van der Meij 2017; Wilcox and Keselman 2003). Note that journal articles increasingly have a “Robustness” section following the main analyses to confirm results using complementary data or models. In *those* contexts, “robust” simply means “hardy” and could be replaced with “replication” or “confirmation testing.” In this research, “robust” is used in the classic sense; it refers to the technical statistical literature on data that do not conform to the normal distribution assumption, typically on the model’s error terms, and nonparametric methods which examine the extent to which a model’s findings will be incorrect if the data do not follow the assumed distributional form. One recommendation is to transform the data to appear more normal-like (cf., Ratcliff 1993). Another recommendation is to use robust statistics—indices that are less influenced by the inclusion of nonnormal or extreme data. Neither of these solutions is perfect, as we show next. The transformations rarely suffice, and as described shortly, the nonparametric or robust statistics are infrequently utilized, so readers might not be able to confidently assess results.

Natural log and square root transformations are often suggested to reduce excessively high data values. The transformations do so, but perhaps not as much as one would like, albeit in theory producing data that more closely approximating normality (Games and Lucas 1966;

Rasmussen 1989; Tukey 1962). For example, we took the distributions from Figure A3 that were skewed at 0.5, 0.75, and 1.0 to explore these transformations. Applying a square root transformation (e.g., $x_i^* = \sqrt{5 + x_i}$) lessened the skewness, but only slightly, to 0.31, 0.63, and 0.87. Applying a natural log transformation (e.g., $x_i^* = \ln(5 + x_i)$) fared a little better, yielding 0.06, 0.44, and 0.69. These transformations seem suboptimal in only partially correcting the nonnormality, and unfortunately, there are no agreed upon standard transformations for data displaying positive or negative kurtosis (Andersen 2008; K. D. Lawrence 2020). Finally, of course, the data must be transformed back to its original scale before results on means or coefficients are interpreted.

The second alternative treatment of nonnormal data is to use non-parametric or robust statistics. These statistics are constructed to be relatively insensitive to small changes in the data with respect to the statistical models employed (Fabozzi et al. 2014, 405). For example, as measures of central tendency, the mean is known to be a non-robust estimator in that it changes in the presence of outlying data points, and by comparison, the median is more robust (Daszykowski et al. 2007; Pollet and van der Meij 2017; Y. Yuan and MacKinnon 2014). As an illustration, consider the previous small data sets (1, 2, 3, 4, 5) and (1, 2, 3, 4, 10). The median equals 3 for both, whereas the mean changed from 3 to 4. While it has been long known that these measures of central tendency convey these properties (i.e., the median being less affected by extreme observations than the mean), the reader can attest to the relatively infrequent use of medians over means in the literature (in part attributable to their lesser statistical efficiencies regarding standard errors, e.g., King and Roberts 2015).

In addition to the median, other robust indices include the non-parametric Mann–Whitney Wilcoxon test (based on ranks, Bakker and Wicherts 2014; Ruscio and Mullen 2012), trimmed means, Winsorized means, and M- or W-estimators, among others (Boos and Stefanski 2015; Huber 1964; Sharma, Durvasula, and Dillon 1989). For trimmed means, the top several and bottom several observations are deleted and the mean is reported based on the remaining data points (the data set has thus been trimmed). Note of course that just as researchers are uncertain as to whether a few outliers may be deleted, the analogous question here is how many data points may be trimmed from the top and bottom before raising similar concerns. For Winsorized means, the top value is replaced by a duplicate of the second value, and the bottom value is replaced by a duplicate of the second to last value, thus bringing in the extremes (e.g., $\frac{1}{7}(x_2 + x_2 + x_3 + x_4 + x_5 + x_6 + x_6)$). However, again, even for that simple adjustment, there are no norms or demonstrated statistical efficacy for how many low and high observations should be reined in (e.g., Yu et al. 2019). For M-estimators (based on maximum likelihood) or W-estimators (based on differentially weighting data points), the calculations can be more complex (e.g., $\frac{1}{n} \sum \psi(x_i - \theta) = 0$, defining the estimator for a robust mean, or $\sum \psi(y_{ij} - b_j x_{ij}) = 0$, for robust regression, with varying ψ scoring functions) (cf., Barbato et al. 2011; Barnett and Lewis 1994; Daszykowski et al. 2007; Hettmansperger and McKean 2010; Huber and Ronchetti 2009; Wilcox 2021).

It is rare to see any of these robust statistics reported in published articles, as readers can judge for themselves. Even when they are, reviewers have little guidance and few norms (e.g., which transformations may be optimal, what constitutes the proper percent of data points to Winsorize, etc.). Given the questions and infrequent utilization of these techniques, there naturally may be confusion regarding the interpretation of such results. In addition, these statistics are known to be less statistically efficient, that is, generally associated with greater standard errors, producing wider confidence intervals, albeit thereby conservative (valued by journals) in being less likely to produce significant results, and important to establish that a result is “real” before it is published.

In addition to robust measures of central tendency, robust indices of dispersion have also been offered. These indices are especially challenging because the sample variance is even more sensitive to outliers

than the sample mean (due to it being squared). It is important to note that researchers should not delete outliers if the focal statistical test is about variance because doing so would directly change the statistic of interest (AMC 1989; Cribbie et al. 2012), and the interquartile range is suggested instead (Pollet and van der Meij 2017).

There are also practical and simple alternatives. For example, cleaning a data set by deleting an outlier observation can be defended as a conservative move because the reduction of the sample size, even just by one, reduces the statistical power. On the other hand, retaining the outlying data point increases the error variance, hence decreasing power for most statistical tests, again, a defensibly conservative move. As a third option, results may be reported that include and exclude any outlying data points (Pollet and van der Meij 2017). This seems like excellent advice for transparency and thoroughness, yet some researchers report that they investigated their findings with and without outliers and that results were “similar” without fully documenting the two variants of data sets or their respective analyses, and if the results were similar, the default should probably be to report the analyses based on the full data set, rather than on any subset thereof.

These various solutions to data that do not seem to reflection the assumed normal distribution, whether due to skewness, kurtosis, or outliers, seem suboptimal, for example, transformations do not yield perfectly clean normal data, alternative statistics are not well-understood in terms of their respective norms for calculation and reporting, more sophisticated alternative statistics are simply not utilized frequently, there too perhaps due to lack of norms for usage and results. Thus, our research expands beyond the statistical literature’s suggestions regarding solutions to data that exhibit skewness, kurtosis, and outliers. That is, we know *theoretically* that nonnormal data may be a problem. Our research focuses on the empirical outcomes and is intended to contribute to the literature by posing the simple, practical question, “If my data look unusual or nonnormal, do I need to do anything, or will my results remain the same?” We answer this question by presenting three sets of simulation studies—to examine nonnormal data in the form of skewness, kurtosis, and the presence of outliers, and their influence on correlations and regressions.